

JKBOSE 2026

CLASS 11TH

RESPIRATION IN PLANTS

LECTURE-05

BOTANY

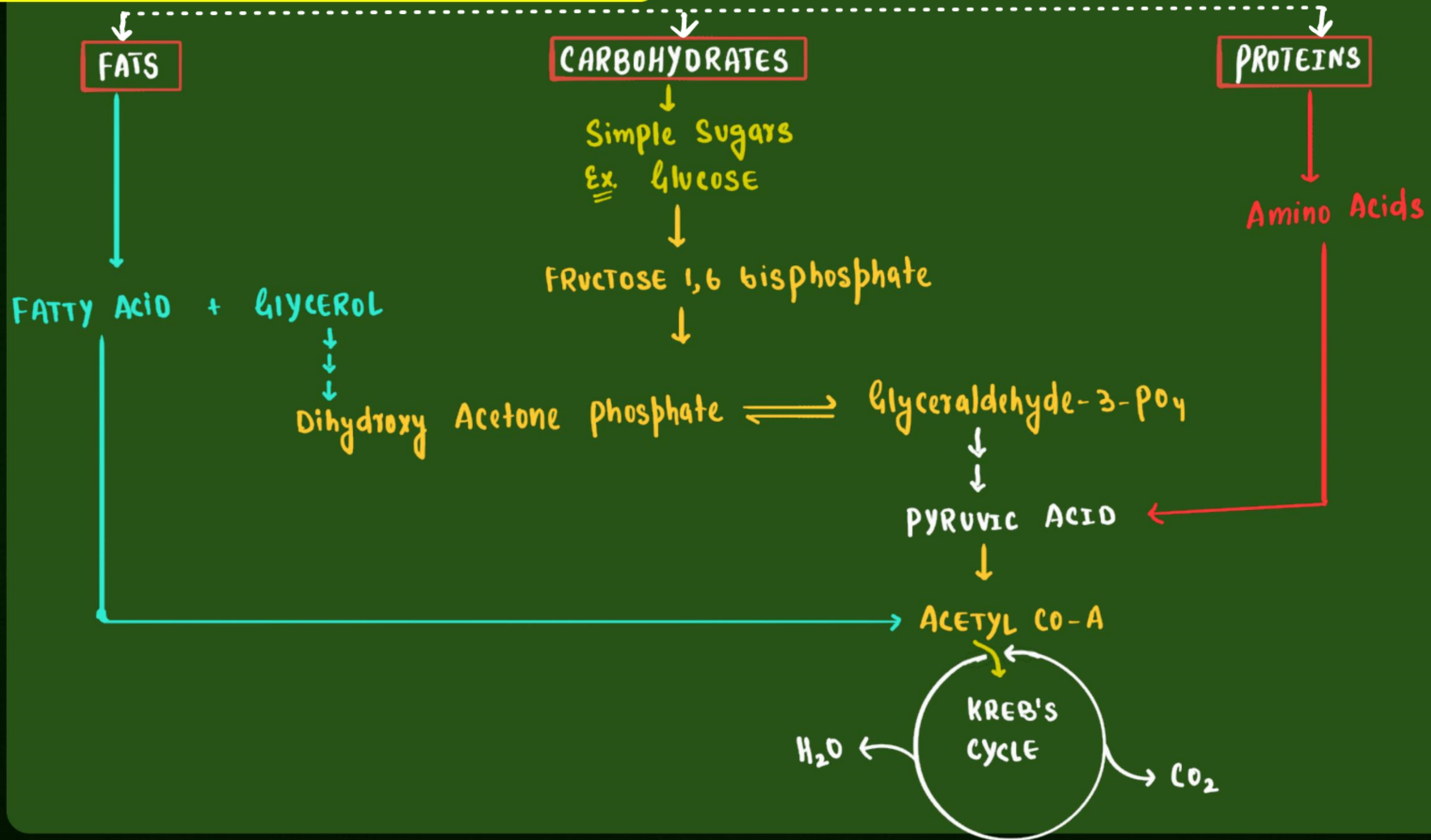
ARSHID BHAT SIR



Topics To Covered

- 1** Amphibolic pathway
- 2** Respiratory Quotient (R.Q).

Amphibolic Pathway



- Most Common Respiratory Substrate are Carbohydrates (ex. Glucose).
- Traditionally we call respiratory pathway as Catabolic pathway because it involves breakdown of respiratory Substrates. But it is wrong.

Q. why?

Reason:

Example: → Fats are broken down into Fatty acids & Glycerol which ultimately forms Acetyl Co-A $\rightsquigarrow$ Catabolic process (during energy requirement).
→ During fat formation inside cells, Acetyl Co-A is withdrawn and forms Fat $\rightsquigarrow$ Anabolic process (during Fat requirement).

Conclusion:

- Respiratory pathway involves both Anabolism and Catabolism.

Thus respiratory pathway is **Amphibolic pathway**

RESPIRATORY QUOTIENT

(R.Q)

$\hat{=}$ $R.Q = \frac{\text{Vol. of } CO_2 \text{ Evolved}}{\text{Vol. of } O_2 \text{ Consumed}}$ over a period of time during respiration.

Examples:

① R.Q of Carbohydrate (ex. $C_6H_{12}O_6$)



$$R.Q = \frac{CO_2}{O_2} = \frac{6}{6} = 1$$

② R.Q of Fat (ex. tripalmitin fat)

$$R.Q = 0.7$$

③ $R.Q \text{ (proteins)} = 0.9$

Note: $R.Q \text{ (Fats \& Proteins)} < 1$

④ R.Q of Anaerobic respiration;

$$R.Q = \frac{CO_2}{\text{zero}} = \infty$$

Significance of R.O :-

- ① It predicts type of respiratory substrate used.
- ② It predicts type of respiration (aerobic / anaerobic)

QUESTIONS

Respiratory Quotient (RQ) value of tripalmitin is

[NEET-2019]

A 0.9 $\rightsquigarrow$ Protein

☒ **B** 0.7 $\rightarrow$ Tripalmitin fat

C 0.07 $\times$

D 0.09 $\times$

HOMEWORK



Revision



*Complete
chapter*

Thank
You